

SOFTWARE LICENSING

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Objectives

- Define and categorize the term software.
- List the ways of obtaining a software.
- Discuss whether a business organization build or buy a software.
- Define the term open-source and identify its primary characteristics.
- Access software and databases from the Internet via a web browser through cloud computing which is referred to as “Software as a service (SaaS)”.



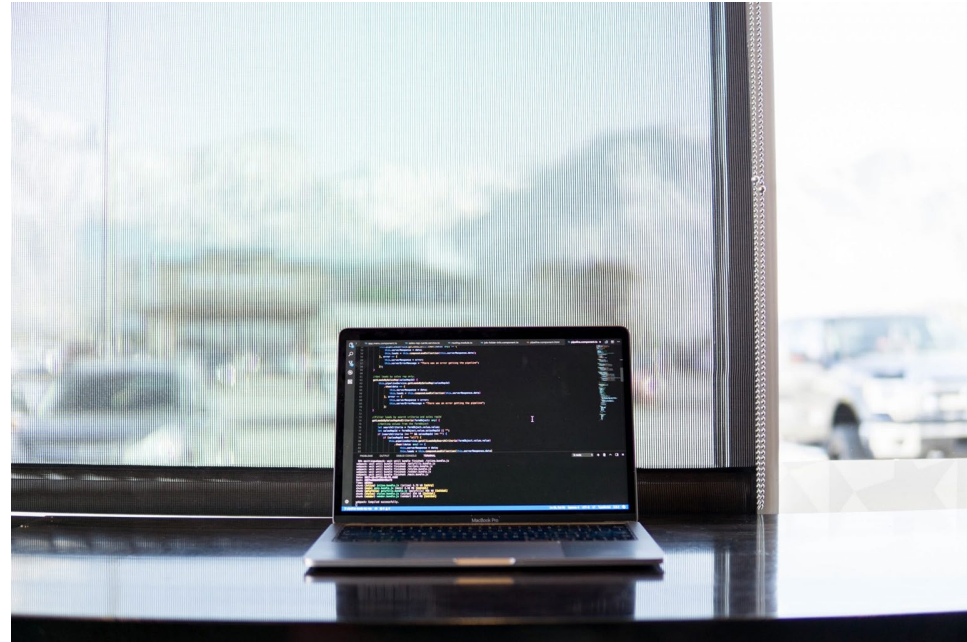
Software Programming

Programming is the process of creating a set of logical instructions for a digital device to follow using a programming language. The process of programming is sometimes called “coding” because the developer takes the design and encodes it into a programming language which then runs on the computer.

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Software Introduction

Software can be broadly divided into two categories:
Operating Systems
and
Application Software.



Operating Systems

Operating systems manage the hardware and create the interface between the hardware and the user.

Key functions:

1. Managing the hardware resources of the computer
2. Providing the user-interface components
3. Providing a platform for software developers to write applications.



Application Software

Application Software is used to accomplish a specific goal such as word processing, calculations on a spreadsheet, or surfing the internet using a web browser.

What are some of the common types of application software that you use?



Application Software Cont...

Productivity Software

- **Word processing**
 - Users can create and edit documents using this class of software.
- **Spreadsheet**
 - This class of software provides a way to do numeric calculations and analysis, displaying the result in charts and graphs.
- **Presentation**
 - Users can create slideshow presentations using this class of software.



Application Software Cont...

Productivity Software

- **Database management system**
 - This software serves as an electronic filing cabinet for records such as customer lists, employee data, and inventory information. Data can be sorted, manipulated and queried to create reports.
- **Desktop publishing**
 - This software combines word processing, graphics, and page layout software to create documents. Allows companies to design and produce sales brochures, catalogs, advertisements, and newsletters in-house.
- **Financial software**
 - This software is used to compile accounting and financial data and create financial statements and reports.



Desktop Software

Desktop software

- applications installed on a personal computer—your browser, your Office suite, photo editors, and computer games are all desktop software.



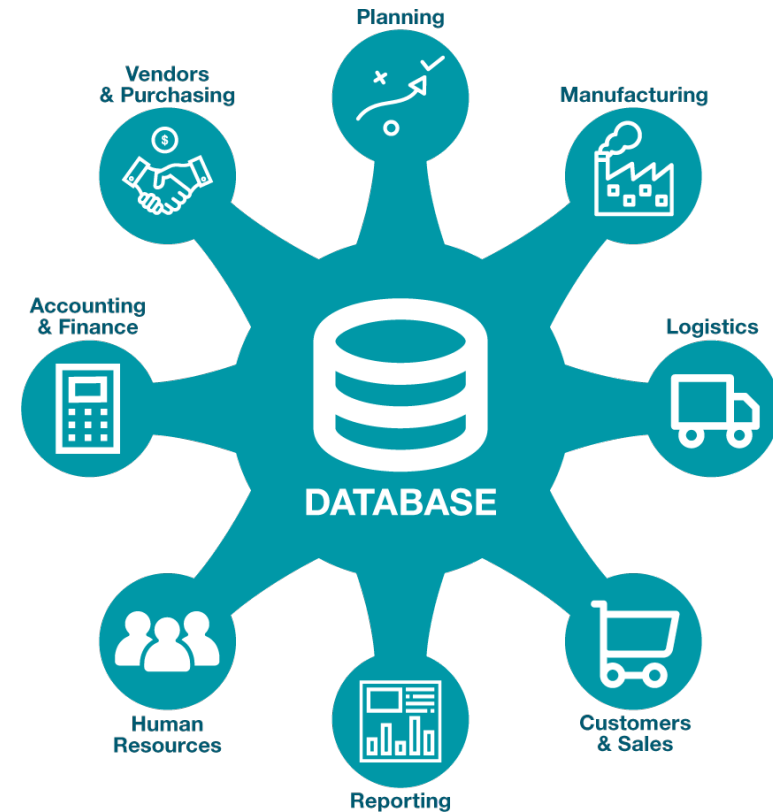
Enterprise Software

Enterprise software

- applications that address the needs of multiple, simultaneous users in an organization or work group.

Enterprise resource planning (ERP)

- software packages contain integrated modules that work together to bring the functions of an entire organization into one program.



Mobile Software

Just as with the personal computer, mobile devices such as **smartphones** and **tablets** also have operating systems and application software. These mobile devices are in many ways just smaller versions of personal computers.



Obtaining Software

Software can be obtained in a number of different ways.

It can be developed internally which is referred to as **proprietary software**.

It can be purchased **off-the-shelf**, also referred to as packaged software.

What are some of the advantages and disadvantages associated with each method?



Build Vs. Buy

When an organization decides that a new software program needs to be developed, they must determine if it makes more sense to build it themselves or to purchase it from an outside company. This is the “build vs. buy” decision.



Build Vs. Buy

Advantages

There are many advantages to purchasing software from an outside company.

- First, it is generally less expensive to purchase software than to build it.
- Second, when software is purchased, it is available much more quickly than if the package is built in-house.



Build Vs. Buy Cont.

Disadvantages

- First, the same software you are using can be used by your competitors.
- Second, if you purchase software from a vendor and then customize it, you will have to manage those customizations every time the vendor provides an upgrade.



Implementation And Maintenance

Once a new software is developed or purchased for a system, the organization must determine the best method for implementation. Convincing a group of people to learn and use a new system can be a very difficult process.

There are several different methodologies an organization can adopt to implement the software for a new system. Four of the most popular are:

- **Direct Cutover**
- **Pilot Implementation**
- **Parallel Operation**
- **Phased Implementation**



Open Source Software (1 of 3)

- An alternative to traditional software development
- Many users and coders can study, share, and modify the code
- Program modifications must be shared with all people on the project and all licenses adhered to



Open Source Software (2 of 3)

- Categorized open source communities into four community types
 - Ad hoc
 - Standardized
 - Organized
 - Commercial



Open Source Software (3 of 3)

- Six different dimensions
 - General structure
 - Environment
 - Goals
 - Methods
 - User community
 - Licensing



Open Source Software

Advantages

- The software is available for free
- The software source-code is available; it can be examined and reviewed before it is installed.
- The large community of programmers who work on open-source projects leads to quick bug-fixing and feature additions.

Disadvantages

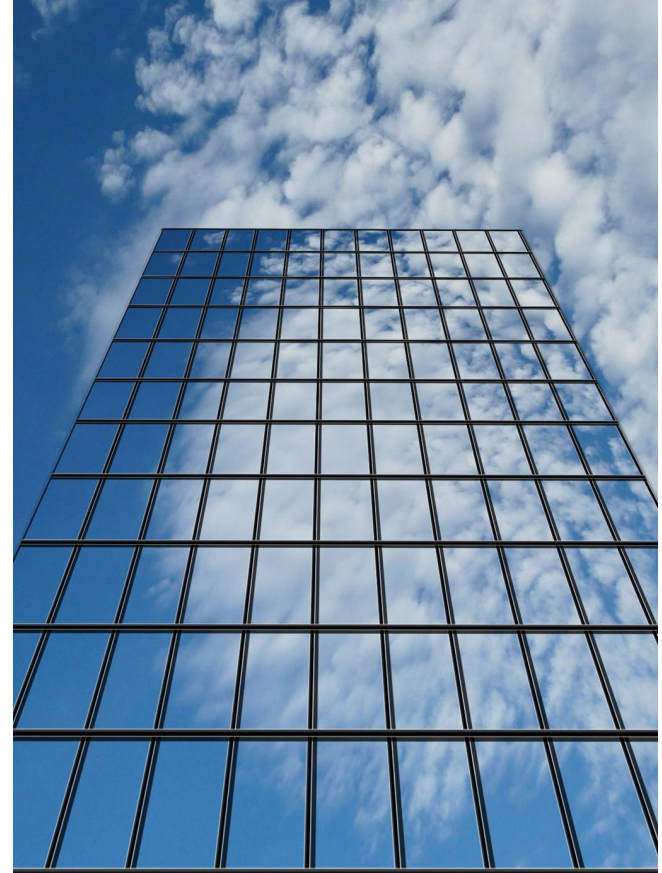
- Technical support from the company that developed the software.
- Companies provide updates and features as required by the users.



Cloud Computing

Through cloud computing applications, services, and data storage are made accessible through the internet.

Software as a service (SaaS) is software that is rented rather than purchased.



Key Takeaways

- There are two basic categories of software: operating systems and applications.
- Desktop applications are typically designed for a single user. Enterprise software supports multiple users in an organization or work group.
- Cloud computing provides access to software and databases from the Internet via a web browser.
- The software as a service (SaaS) category of cloud computing gives companies access to a large assortment of software packages without having to invest in hardware or install and maintain software on its own computers.



Key Takeaways

- Software development is about solving business problems.
- Developing new software applications requires several steps, from the formal SDLC process to more informal processes such as agile programming or lean methodologies.
- In end-user computing, software development happens outside the information technology department.
- Conducting a feasibility analysis is important to ensure the new system will be a good investment.



Key Takeaways

- Programming languages have evolved from very low-level machine-specific languages to higher-level languages that allow a programmer to write software for a wide variety of machines.
- Most programmers work with software development tools to make the software development more efficient.
- Some companies choose to purchase software to save development costs and speed implementation.
- When implementing new software applications, there are several different types of implementation methodologies that must be considered.



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